



Bhabha University, Bhopal (M.P.)

Scheme of Examination

Program – Bachelor Of Technology

Branch – Mechanical Engineering

Session: 2022-2023

Eighth Semester

Sr. No.	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory Slot)			Marks for End Sem. Examination		Total Marks
			L	T	P		End Sem. Exam.	Tests (Two)	Assign ments /Quiz	End Sem. Practic al /Viva	Practical Record/Assi gnment/Qui z /Presentation	
1	ME-81	Refrigeration & Air Conditioning	3	1	2	5	70	20	10	30	20	150
2	ME-82	Alternative Automotive Fuels & Emissions	3	1	2	5	70	20	10	30	20	150
3	ME-83	Energy Management & Audit	3	-	-	3	70	20	10	-	-	100
4	ME-84	Maintenance Management	3	-	-	3	70	20	10	-	-	100
5	ME-85	Major Project	-	-	6	6	-	-	-	120	80	200
6	ME-86	Seminar & Group Discussions(Only Internal Assessment)	-	-	4	2	-	-	-	-	50	50
		Total	12	2	14	24	280	80	40	180	170	750

L: Lecture T: Tutorial P: Practical

**Note: End semester Examination
will be of 3 hrs.**



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Mechanical Engineering, VIII-Semester

ME-81 Refrigeration & Air Conditioning

Unit-I Introduction: Principles and methods of refrigeration, freezing; mixture cooling by gas reversible expansion, throttling, evaporation, Joule Thomson effect and reverse Carnot cycle; unit of refrigeration, coefficient of performance, vortex tube & thermoelectric refrigeration, adiabatic demagnetization; air refrigeration cycles- Joule's cycle Boot-strap cycle, reduced ambient cycle and regenerative cooling cycles.

Unit-II Vapor compression system: Vapor compression cycle, p-h and t-s diagrams, deviations from theoretical cycle, sub-cooling and super heating, effects of condenser and evaporator pressure on cop; multi-pressure system: removal of flash gas, multiple expansion & compression with flash inter cooling; low temperature refrigeration: production of low temperatures, cascade system, dry ice, production of dry ice, air liquefaction system,.

Unit-III (a) Vapor absorption system: Theoretical and practical systems such as aqua-ammonia, electrolux & other systems; (b) **Steam jet refrigeration:** Principles and working, simple cycle of operation, description and working of simple system, (c) **refrigerants:** nomenclature & classification, desirable properties, common refrigeration, comparative study, leak detection methods, environment friendly refrigerants and refrigerant mixtures, brine and its properties

Unit-IV Psychometric: Calculation of psychometric properties of air by table and charts; psychometric processes: sensible heating and cooling, evaporative cooling, cooling and dehumidification, heating and humidification, mixing of air stream, sensible heat factor; principle of air conditioning, requirements of comfort air conditioning, ventilation standards, infiltrated air load, fresh air load human comfort, effective temperature & chart, heat production & regulation of human body,



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Mechanical Engineering, VIII-Semester

ME-81 Refrigeration & Air Conditioning

Unit-V Air conditioning loads: calculation of summer & winter air conditioning load, bypass factor of coil, calculation of supply air rate & its condition, room sensible heat factor, grand sensible heat factor, effective sensible heat factor, dehumidified air quantity. Problems on cooling load calculation. Air distribution and ventilation systems

References:

1. Arora CP; Refrigeration and Air Conditioning; TMH
2. Sapali SN; Refrigeration and Air Conditioning; PHI
3. Ananthanarayan; Basic Refrigeration and Air conditioning; TMH
4. Manohar Prasad; Refrigeration and Air Conditioning; New Age Pub
5. Ameen; Refrigeration and Air Conditioning; PHI
6. Pita ; Air conditioning Principles and systems: an energy approach; PHI
7. Stoecker W.F, Jones J; Refrigeration and Air conditioning; McGH, Singapore
8. Jordan RC and Priester GB Refrigeration and Air Conditioning, PHI USA
9. Arora RC; Refrigeration and Air conditioning; PHI Learning

List of Experiments :-

1. General Study of vapor compression refrigeration system.
2. General Study of Ice Plant
3. General Study and working of cold storage
4. General Study Trane Air Condition (Package Type).
5. General Study of Electrolux Refrigeration
6. General Study One tone thermax refrigeration unit.
7. General Study of Water cooler
8. General Study of Psychrometers (Absorption type)
9. General Study of Leak Detectors (Halide Torch).
10. General Study and working of Gas charging Rig.
11. General Study of window Air Conditioner.
12. General Study and working of Vapor compression Air conditioning Test rig.
13. Experimentation on Cold Storage of Calculate COP & Heat Loss.
14. Experimentation on Vapor compression Air Conditioning test rig.
15. Changing of Refrigerant by using Gas Charging Kit.



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Mechanical Engineering, VIII-Semester

ME-82 Alternative Automotive Fuels & Emissions

Unit 1: Introduction Automobile Fuels:

Classification of Automobile alternative fuels(liquid, gaseous, hydrogen, LPG, CNG, Biogas etc.), Desirable characteristics of SI & CI engine alternative fuels, Rating of SI & CI engine fuels, Introduction to alternate energy sources. Like EV, hybrid, fuel cell and solar cars. merits and demerits of various alternate fuels.

Unit 2: Liquid alternative fuels:

Vegetable Oils: Various vegetable oils for automobile engines, esterification, performance in engines, performance and emission characteristics, bio diesel and its characteristics. Alcohols: Properties as engine fuel, alcohols and gasoline blends, performance in automobile engine, methanol and gasoline blends.

Unit 3: Gaseous Fuels:

Biogas: Introduction to Biogas system, Process during gas formation, Factors affecting biogas formation. Usage of Biogas in SI engine & CI engine., Properties of Natural gas, Hydrogen gas, LPG & CNG as engine fuels, storage and handling, performance and safety aspects to all gaseous fuel, fuel metering systems.

Unit 4: Automobile emissions:

Types of automobile emissions, emission characteristics, formation of automobile emissions, mechanism of HC , CO and NO in SI engine, exhaust emission and factors affecting the emission, evaporative emission, crankcase emission, lead emission CI engine emissions: formation of smoke, factors affecting the smoke formation, unburned hydrocarbons, carbon monoxide, oxides of nitrogen, smog and comparison of diesel and petrol emissions.

Unit 5:Emissions Norms & Measurement:

Emission norms as per Bharat Standard up to BS – IV and procedures for confirmation on production. Demerits of automobile emission to environment. Types Of Catalytic Conversion, Measurement Techniques Emission Standards and Test Procedure NDIR, Gas Chromatograph, smoke meters, emission standards.



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Mechanical Engineering, VIII-Semester

ME-82 Alternative Automotive Fuels & Emissions

References:

1. J.B. Heywood. Internal combustion Engines, Wiley
2. Ganeshan V; Internal Combustion engines; TMH
3. Mathur M L & Sharma RP; A. Course in IC engines; Dhanpat Rai
4. R Yadav, Internal Combustion Engines
- 5 Halderman JD and Mitchell CD; Automotive Engines theory and servicing; Pearson
6. DomKundwar; Internal Combustion Engines; Dhanpat Rai Publications
7. Taylor GF; Internal Combustion Engines Theory & Practice; MIT Press
8. Richard Stone; Introduction to IC Engines; Society of Automotive Engr (Palgrave Mc Millan)

List of Experiment (Pl. expand it):

1. Study of alternative fuel for automobile.
2. Study of esterification of alternative fuels.
3. Study of blending different types of bio-diesel.
4. Measurement of smoke from automobile.
5. Study of different types of emissions.
6. Study of various techniques for NO_x reduction



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Mechanical Engineering, VIII-Semester

ME-83 Energy Management & Audit

UNIT-I Energy Management: Concept of energy management, energy demand and supply, economic analysis; Duties and responsibilities of energy managers.

Energy Conservation: Basic concept, energy conservation in Household, Transportation, Agricultural, service and Industrial sectors, Lighting, HAVC.

UNIT-II Energy Audit: Definition, need and types of energy audit; Energy management (Audit) approach: Understanding energy cost, bench marking, energy performance, matching energy use to requirement, maximizing system efficiencies, optimizing the input energy requirement; Fuel & energy substitution; Energy audit instruments; Energy conservation Act; Duties and responsibilities of energy manager and auditors.

UNIT-III Material energy balance: Facility as an energy system; Method for preparing process flow; material and energy balance diagrams.

Energy Action Planning: Key elements, force field analysis; Energy policy purpose, perspective, content, formulation, rectification

UNIT-IV Monitoring and Targeting: Definition monitoring & targeting; Data and information analysis.

Electrical Energy Management: energy conservation in motors, pumps and fan systems; energy efficient motors.

UNIT-V Thermal energy management: Energy conservation in boilers, steam turbine and industrial heating system; Application of FBC; Cogeneration and waste heat recovery; Thermal insulation; Heat exchangers and heat pump; Building Energy Management.



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Mechanical Engineering, VIII-Semester

ME-83 Energy Management & Audit

References:

1. Murphy & Mckay, Energy Management, BSP Books Pvt. Ltd.
2. Smith CB; Energy Management Principle, Pergamon Press, New York.
3. Rajan GG, Optimising Energy Efficiency in Industry, TMH.
4. Callaghan P O, Energy Management, McGraw-Hill Book Company.
5. Amit Kumar Tyagi, Handbook on Energy Audit and Management, Tata Energy Research Institute.
6. Bureau of Energy Efficiency, Study material for energy Managers and Auditors: Paper I to V.
7. Hamies; Energy Auditing and Conservation: Method, Measurement□, Hemisphere, Washington.
8. Witty, Larry C, Industrial Enegy Management Utilisation, Hemisphere Publishers, Washington
9. Kreith & Goswami, Energy Management and Conservation Handbook, CRC Press



Bhabha University, Bhopal (M.P.)

Mechanical Engineering, VIII-Semester

ME-84 Maintenance Management

UNIT-I Maintenance, definition, preventive, corrective, on- line off- line maintenance, window maintenance, emergency, reconditioning, design out maintenance . “ product- item- machine plant structure characteristics. Design , cost and safety aspects.

UNIT-II Production- maintenance system, Maintainability, Maintenance procedures, guidelines for matching procedures to items, universal maintenance procedures. Shutdown programs

UNIT-III Maintenance organization, work load, resource characteristics, administrative structure, work planning, scheduling and control strategy, feed back, combinations of manpower, tools and spares. Documentations. Network planning, computer based management information systems,.

UNIT-IV restoration of components, assembly, disassembly bush bearing, housings, Ball and roller bearings, key-splines, couplings shafts- lead screw fittings, clutches- brakes, belt pulley, chain sprocket, guideways, machine hydraulics, pneumatics, electrical works and motors, seals, and packings. Fasteners , welding, machining, repair cycles, repair complexities, maintenance stages. Lubrication, accuracies and technological test charts.

UNIT-V Failure statistics, Failure data, failure patterns/ statistical models, Failure analysis, applications of different models, Depreciation and average machine life. case studies.

Books:

- 1 .Maintenance management – Hand book, Higgins
- 2.Maintenance planning and control, Anthony Kelly



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ME-84 Maintenance Management

Tutorial topics.

- [1] Drawings, manuals and specifications.
- [2] Limits, fits and tolerances, Surface finish
- [3] Measurement tools
- [4] Engineering materials
- [5] Wear mechanisms
- [6] Maintenance softwares
- [7] Industrial safety norms and regulations
- [8] Motivation



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Mechanical Engineering, VIII-Semester

ME-85 Major Project

Objectives of the course Minor/Major Project are:

- ☐ To provide students with a comprehensive experience for applying the knowledge gained so far by studying various courses.
- ☐ To develop an inquiring aptitude and build confidence among students by working on solutions of small industrial problems.
- ☐ To give students an opportunity to do some thing creative and to assimilate real life work situation in institution.
- ☐ To adapt students for latest development and to handle independently new situations.
- ☐ To develop good expressions power and presentation abilities in students.

The focus of the Major Project is on preparing a working system or some design or understanding of a complex system using system analysis tools and submit it the same in the form of a write up i.e. detail project report. The student should select some real life problems for their project and maintain proper documentation of different stages of project such as need analysis market analysis, concept evaluation, requirement specification, objectives, work plan, analysis, design, implementation and test plan. Each student is required to prepare a project report and present the same at the final examination with a demonstration of the working system (if any)

Working schedule The faculty and student should work according to following schedule: Each student undertakes substantial and individual project in an approved area of the subject and supervised by a member of staff. The student must submit outline and action plan for the project execution (time schedule) and the same be approved by the concerned faculty. Action plan for Major Project work and its evaluation scheme #(Suggestive)



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Mechanical Engineering, VIII-Semester

ME-85 Major Project

Task/Process	Week	Evaluation	Marks For Term Work#
Orientation of students by HOD/Project Guide	1st	-	-
Literature survey and resource collection	2nd	-	-
Selection and finalization of topic before a committee	3rd Seminar	I	10
Detailing and preparation of Project (Modeling, Analysis and Design of Project work	4th to 5th	-	10
Development stage		-	
Testing, improvements, quality control of	6th to 10th 11th	-	25
Acceptance testing	12th	-	10
Report Writing	13th to 15th	-	15
Presentation before a committee (including user manual, if any)	16th	Seminar-II	30



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* Committee comprises of HOD, all project supervisions including external guide from industry (if any)

The above marking scheme is suggestive, it can be changed to alternative scheme depending on the type of project, but the alternative scheme should be prepared in advance while finalizing the topic of project before a committee and explained to the concerned student as well.

NOTE: At every stage of action plan, students must submit a write up to the concerned guide